

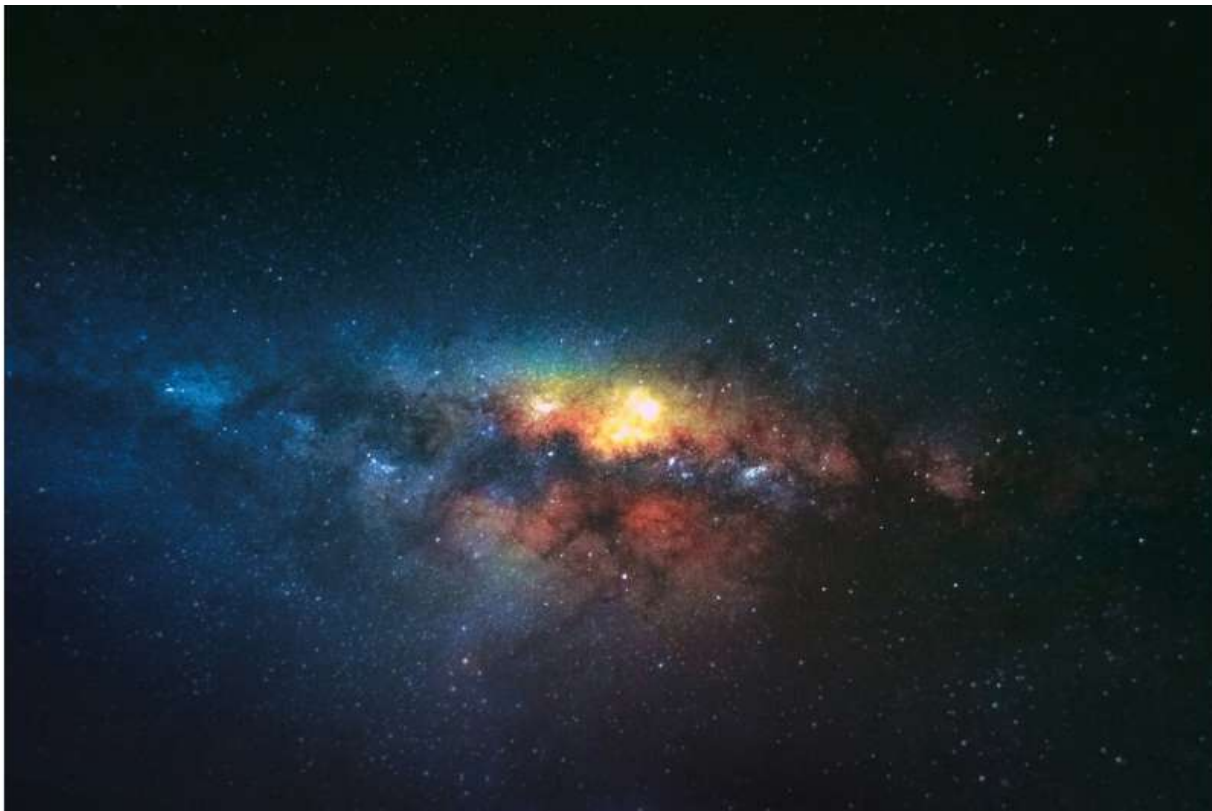
[Home](#) / [Astronomy & Space](#) / [Astronomy](#)

 MAY 28, 2026

Taking dark energy out of the equation: Mathematicians challenge the standard cosmological model of the universe

by [Greg Watry, UC Davis](#)

edited by [Sadie Harley](#), reviewed by [Robert Egan](#)



Credit: [Unsplash/CC0 Public Domain](#)

Mathematicians are challenging the idea that dark energy is responsible for

the accelerating expansion of the universe. In a [new paper published in *Proceedings of the Royal Society A*](#), mathematicians from the University of California, Davis, provide mathematical proof that instabilities inherent in the Einstein-Euler equations imply that the current model of the expanding universe is not viable.

The Einstein-Euler equations are a union of general relativity and fluid dynamics equations used to model astronomical phenomena such as galaxies, black holes, and cosmic expansion.

The research directly challenges the [Lambda-cold dark matter model](#), the standard cosmological model of the Big Bang.

Study corresponding author Blake Temple, a distinguished professor emeritus of mathematics at UC Davis, compared the standard cosmological model to a pencil standing on its tip.

"All the forces are in balance when a pencil is standing on end, so it is a 'solution of the equations,'" he said. "But it's unstable. Any breath of air and it falls away."

The mathematics, Temple said, prove that Friedmann spacetimes—mathematical models that govern cosmic expansion—are unstable at both small and large length scales at the Big Bang, making it the most unstable solution of all.

"Unstable solutions in physics and science are considered not physical," Temple said. "You'll never observe them in nature."

Temple noted that this instability suggests a simpler explanation—one based entirely within the framework of Einstein's original theory.

"The instability of all Friedmann spacetimes to accelerated expansion suggests a simpler, more natural explanation for the acceleration of the universe than dark energy," he said.

Explaining the universe's accelerating expansion

Almost 30 years ago, dark energy was proposed as the force responsible for the accelerating expansion of the universe.

The idea harkens back to Albert Einstein's original 1915, gravity-describing equations for general relativity. To produce a static universe, Einstein initially introduced an antigravity factor in his theory. He called this factor the "cosmological constant."

After Edwin Hubble discovered the universe was expanding in 1929, Einstein famously called the cosmological constant his "biggest blunder" because without it he could have predicted the expansion.

However, the cosmological constant, and the idea that it's interchangeable with dark energy, was reintroduced to explain the universe's accelerating expansion in the 1990s. Standard cosmological models are based on what's called the "Friedmann universe," which describes all matter as expanding but being evenly distributed throughout space at each fixed time.

But the math didn't add up to Temple and his colleagues, leading them to pursue alternative explanations for the accelerating expansion of the universe.

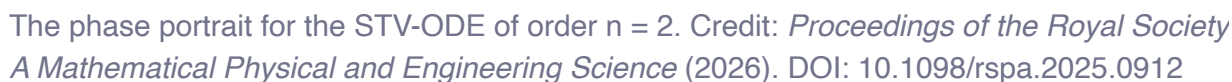
"Our first idea was that maybe the universe was expanding because there was a shockwave, and the anomalous acceleration was the expanding wave behind that shockwave," Temple said. "Then we realized there's a family of self-similar solutions during the radiation epoch of the Big Bang, which might model that expanding wave."

Self-similar equations describe physical phenomena that maintain a pattern or structure, regardless of its scale.

In the current paper, the mathematicians use a self-similar version of the Einstein equations, which they derived in prior work, to represent the

"We prove that, like Einstein's static model, the Friedmann spacetimes are all unstable to radial perturbation at large length scales," Temple said. "This appears to rule out the Lambda-cold dark matter model as a viable stable solution of the Einstein equations of general relativity, with or without dark energy."

"This means," he added, "that the Big Bang should generically look exactly like a Friedmann spacetime near the center of symmetry, but generically one should observe accelerations away from Friedmann far from the center."



Rethinking the Copernican principle?

Temple and his colleagues found that the accelerating expansion of the universe is a direct consequence of the Einstein-Euler equations without the insertion of a cosmological constant or dark energy.

The math also calls into question the Copernican principle—the idea that Earth's location does not occupy a special place in the universe.

"Both the Lambda-cold dark matter model and a spherically symmetric spacetime produce a special place where we must lie for the model to be physically plausible," Temple said. "If this principle rules out one, it has to rule out the other."

Publication details

C. Alexander et al, The instability of critical and underdense Friedmann spacetimes at the Big Bang as an alternative to dark energy, *Proceedings of the Royal Society A Mathematical Physical and Engineering Science* (2026). DOI: [10.1098/rspa.2025.0912](https://doi.org/10.1098/rspa.2025.0912)

Journal information: [Proceedings of the Royal Society A](#)

Key concepts

[Big Bang theory](#)[Expanding universe](#)[Accelerating universe](#)[Cosmological constant](#)

Provided by [UC Davis](#)

Citation: Taking dark energy out of the equation: Mathematicians challenge the standard cosmological model of the universe (2026, May 28) retrieved 2 June 2026 from <https://phys.org/news/2026-05-dark-energy-equation-mathematicians-standard.html>

This document is subject to copyright. Apart from any fair dealing for the purpose of private study or research, no part may be reproduced without the written permission. The content is provided for information purposes only.



PODS big sale is extended! - Loading Without The Rush -

PODS biggest sale of the year is extended — save up to 30% on moving and storage. Moving soon? PODS biggest sale of the year

